



**UNIVERSITY OF  
ILLINOIS PRESS**

---

Proto-Germanic /i/ and /e/ Revisited

Author(s): Fausto Cercignani

Source: *The Journal of English and Germanic Philology*, Vol. 78, No. 1 (Jan., 1979), pp. 72-82

Published by: University of Illinois Press

Stable URL: <https://www.jstor.org/stable/27708429>

Accessed: 18-03-2026 21:35 UTC

---

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact [support@jstor.org](mailto:support@jstor.org).

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



**JSTOR**

*University of Illinois Press* is collaborating with JSTOR to digitize, preserve and extend access to *The Journal of English and Germanic Philology*

## PROTO-GERMANIC \*/i/ AND \*/e/ REVISITED

*Fausto Cercignani, University of Pisa*

Ever since the late 1870's, when the view that Proto-Indo-European exhibited only three vowel qualities (*ī*, *ū*, *ā*) was shown to be incorrect,<sup>1</sup> most Germanists have had to reconcile themselves to the asymmetry which the identification of PIE \*/o/ (and \*/ə/) with \*/a/ produced in the Proto-Germanic short/lax subsystem: two front phonemes (\*/i/ and \*/e/) versus one back phoneme (\*/u/). But while some have presented the Proto-Germanic short/lax vowels in a rectangle which gives a false impression of symmetry,<sup>2</sup> others have attempted to restore a genuine symmetry by assuming that after certain early vowel modifications (or 'umlauts') produced by both vocalic and consonantal influences, Proto-Germanic came to exhibit the parallel phonemes \*/i ~ e/ and \*/u ~ o/.<sup>3</sup>

In a recent contribution on Proto-Germanic morphophonemics, Frans van Coetsem notes that the ensuing controversy has not yet been settled one way or the other and—having lamented “the difficulty of determining in an exact way (1) whether or not there was a PGmc. *u*-umlaut [and] (2) what the changes were in nonstressed position”—concludes that “even if there was *u*-umlaut in PGmc., there may or may not have been total complementarity of *i* and *e*.”<sup>4</sup>

As can be seen, the assumption of a PGmc. \*/i ~ e/ implies the following set of postulates: (1) Proto-Germanic was affected by 'umlaut' phenomena, (2) consequently, pre-Gothic itself was originally affected

<sup>1</sup> See further Eduard Prokosch, *A Comparative Germanic Grammar* (Philadelphia, 1939), § 34a and the references there given.

<sup>2</sup> See, e.g., Winfred P. Lehmann, “The Conservatism of Germanic Phonology,” *JEGP*, 52 (1953), 149, and Hreinn Benediktsson, “The Proto-Germanic Vowel System,” *To Honor Roman Jakobson* (The Hague, 1967), I, 175.

<sup>3</sup> See especially James W. Marchand, “Germanic Short \*i and \*e: Two Phonemes or One?” *Lg.*, 33 (1957), 346–54, and cf. below. Marchand's view has been adopted by William G. Moulton, “Zur Geschichte des deutschen Vokalsystems,” *PBB*, 83 (Tüb., 1961), 6–12; Elmer H. Antonsen, “Germanic Umlaut Anew,” *Lg.*, 37 (1961), 218; and Baldur Panzer, “Die westgermanischen Kurzvokale im Niederdeutschen,” *FoL*, 1 (1967), 251. But Antonsen no longer subscribes to this view—cf. n. 13, below, and see his “On Defining Stages in Prehistoric Germanic,” *Lg.*, 41 (1965), 27, and “The Proto-Germanic Syllabics (Vowels),” *Toward a Grammar of Proto-Germanic*, eds. Frans van Coetsem and Herbert L. Kufner (Tübingen, 1972), pp. 122–23.

<sup>4</sup> See Frans van Coetsem, “Proto-Germanic Morphophonemics,” *Toward a Grammar of Proto-Germanic* (Tübingen, 1972), p. 195.

by these vowel modifications, and (3) the operation of certain changes in unaccented syllables and of certain 'umlaut' phenomena (including an *u*-umlaut of \*/e/ to \*/i/) resulted in total complementarity of PGmc. \*/i/ and \*/e/. It is the purpose of the present study to reassert that an *u*-umlaut of \*/e/ to \*/i/ can be assumed only for the prehistory of Old Saxon and Old High German, and to show that even if the first two postulates mentioned above could be accepted as valid,<sup>5</sup> and the change \*/e/ > \*/i/ before \*[-u] ascribed to Proto-Germanic, it would still be necessary to assume a PGmc. \*/i/ ≠ \*/e/ rather than \*/i ~ e/, since the opposition between \*/i/ and \*/e/ was preserved throughout the prehistory of all the North and West Germanic languages.

The postulation of a PGmc. \*/e ~ i/ appears to have developed from the notion that after the operation of the *i*-umlaut of \*/e/ to \*/i/ (including \*/e/ > \*/i/ before clusters of nasal plus consonant)<sup>6</sup> and of the *a*-umlaut of \*/i/ to \*/e/, the additional change \*/e/ > \*/i/ before \*[-u] produced a total complementarity of \*/i/ and \*/e/ at least in some dialects. W. Freeman Twaddell avoided this conclusion by assuming that the *u*-umlaut of \*/e/ to \*/i/ (as well as the *a*-umlaut of \*/i/ to \*/e/) occurred "only in some cases in some dialects," that is, that it was restricted not only to certain dialects, but also to specific words within those dialects.<sup>7</sup> A few years later, however, J. Fourquet suggested that while in Old English and Old Icelandic the 'progressive neutralization' of the opposition \*/i/ ≠ \*/e/ was incomplete, in Old High German and Old Saxon it reached consummation as a result of the *u*-umlaut of \*/e/ to \*/i/.<sup>8</sup> In the same year, J. Kuryłowicz claimed that "the sporadic 'breaking' of *ī* to *ē* in West and North G., especially in German, and the occasional raise of *ē* to *ī* before *ū* in German and Saxon, furnish a clue to the comprehension of the Gothic vowel-system": since the Gothic distribution presupposes "a previous phonological coincidence of Germanic *ī* and *ē*," similar changes could be assumed for pre-Gothic, though with a subsequent phonetic identification of the allophones \*/i/ and \*/e/ (as well as \*/u/ and \*/o/).<sup>9</sup>

It was apparently as an elaboration of this approach in Proto-Germanic terms that the theory of a PGmc. \*/i ~ e/ was first worked

<sup>5</sup> But see Fausto Cercignani, "Early 'Umlaut' Phenomena in the Germanic Languages," *Lg.*, 56 (1980) and "Alleged Gothic Umlauts," *IF*, 84 (1979).

<sup>6</sup> Cf. Herbert Penzl, *Vom Urgermanischen zum Neuhochdeutschen* (Berlin, 1975), § 4.5a. For \*/e/ > \*/i/ in unaccented syllables see below.

<sup>7</sup> See W. Freeman Twaddell, "The Prehistoric Germanic Short Syllabics," *Lg.*, 24 (1948), 145.

<sup>8</sup> See J. Fourquet, "The Two *e*'s of Middle High German: A Diachronic Phonemic Approach," *Word*, 8 (1952), 130-31.

<sup>9</sup> See J. Kuryłowicz, "The Germanic Vowel System," *BPTJ*, 11 (1952), 53-54.

out by James W. Marchand.<sup>10</sup> Having explained seeming exceptions as due to analogical levelings after the relevant phonemizations,<sup>11</sup> Marchand reasserts that the *a*-umlaut of \*/i/ to \*/e/ cannot be dissociated from the *a*-umlaut of \*/u/ to \*/o/, explicitly ascribes the *u*-umlaut of \*/e/ to \*/i/ to Proto-Germanic—which thus came to exhibit, in his view, the parallel phonemes \*/i ~ e/ and \*/u ~ o/—and concludes that whereas in North and West Germanic the loss of vocalic determinants resulted in the phonemization of PGmc. \*/i/ and \*/e/ (as well as of PGmc. \*/u/ and \*/o/), in pre-Gothic it led to an obliteration of the phonetic difference between PGmc. \*/i/ and \*/e/ (as well as between PGmc. \*/u/ and \*/o/).<sup>12</sup>

Now, quite apart from the validity of the Proto-Germanic umlaut assumption and its implications with regard to pre-Gothic, Marchand's theory can be shown to be untenable on two counts.<sup>13</sup> The first is that the *u*-umlaut of \*/e/ to \*/i/ is typical only of Old Saxon and Old High German, where it is attested before \*/-u/ representing both PGmc. \*/-u/ and PGmc. \*/-ō/: OS OHG *filu* 'much' versus OI *fiql-*, OFris. *fel[o]*, OE *fela/feolu* (Goth. *filu* < \*/felu/) and OS *gibu*, OHG *gibu* 'I give' versus OI *gef*, OFris. *ieve*, OE *giefe/ġeofu* (Goth. *giba* < \*/geþō/).<sup>14</sup> In the other languages there is no conclusive example of

<sup>10</sup> For earlier cursory remarks about the complementarity, or virtual complementarity, of PGmc. \*/i/ and \*/e/ see, e.g., George Trager and Henry L. Smith, "A Chronology of Indo-Hittite," *SIL*, 8 (1950), 67, 69, 70; Antoine Meillet, *Caractères généraux des langues germaniques*, 3rd ed. (Paris, 1926), p. 61; and Prokosch, § 41a, n.

<sup>11</sup> Cf. also James W. Marchand, "The Phonemic Status of OHG *e*" (*Word*, 12 [1956], 84, 86), where Old High German pairs like *gibu* 'I give' – *gebu* 'gift (dat. sg.)' are adduced against Fourquet's postulation of an OHG OS /i ~ e/, and explained as reflecting paradigmatic levelings after the phonemization of pHG \*/i/ and \*/e/ upon the loss of their vocalic determinants.

<sup>12</sup> See Marchand, *Lg.*, 33 (1957), 346–54.

<sup>13</sup> The objection that if \*/i/ and \*/e/ were one phoneme, "there would be no possibility of leveling between these two variants" (Elmer H. Antonsen, "The Proto-Norse Vowel System and the Younger Futhark," *ScS*, 35 [1963], 197–98, n. 8, and cf. his "Zum Umlaut im Deutschen," *PBB*, 86 [Tüb., 1964], 183–84, n. 21) is clearly refuted by analogical levelings between \*/u/ and \*/o/: not only OHG *skif/skef* 'ship,' but also OI *sunr/sonr* 'son'—cf. Marchand, *Lg.*, 33 (1957), 347–48, 350–51, and Hans H. Hock, "On the Phonemic Status of Germanic *e* and *i*," *Issues in Linguistics: Papers in Honor of Henry and Renée Kahane*, ed. B. B. Kachru et al. (Urbana, 1973), p. 327.

<sup>14</sup> Ru. *gibu* for expected *gebu* should be explained as reflecting \*/e/ > \*/i/ before \*/-i/ in the second and third singular—cf. Wolfgang Krause, *Die Sprache der urnordischen Runeninschriften* (Heidelberg, 1971), § 36.2, and Elmer H. Antonsen, *A Concise Grammar of the Older Runic Inscriptions* (Tübingen, 1975), p. 22. The change \*/-ō/ > \*/-ū/ proposed by the American scholar "finally and before a high vowel with or without an intervening \*/n/" (Antonsen, *Toward a Grammar*, p. 139) cannot of course be regarded as late Proto-Germanic but, at most, as 'Northwest Germanic'—cf. Antonsen, "Old High German and the Laws of Final Syllables," *SIL*,

this change, since the unique OI *sīpr*, OE *sidu*/(*sio*)*seodu*, OFris. *side*, OS *sidu*, OHG *situ* (Goth. *sidus*) 'custom' need not be connected with Gk. *ἔθος*,<sup>15</sup> while problematic forms like OFris. *si[u]gun*, OE *sifun*/(*sio*)*seofon* 'seven' (OS *sibun*, OHG *sibun*, Goth. *sibun* < \*/seβun/)<sup>16</sup> and OE *milc*/(*mio*)*meol[u]c* 'milk' (OI *miqlk*, OFris. *melok*, OS *miluk*, OHG *miluh*, Goth. *miluks* < \*/meluks/) may well reflect the change \*/e/ > \*/i/ before \*[-i] in inflected forms exhibiting the sequence \*/eCuCi/,<sup>17</sup> for example, dat. sg. \*/seβuni/ < \*/seβuniji/ < \*/seβuneji/ and gen. sg. \*/melukiz/ < \*/melukez/. Moreover, both Old Icelandic and Old English, as well as Old Frisian, preserve the original opposition \*/i/ ≠ \*/e/ before \*[-u]: OI *liþr* (< \*/liβuz/) – *miqþr* (< /meðuz/), OE *liþu*–*liopu*–*medu/meodu*, OFris. *lith*<sup>18</sup> 'limb' – *mede* 'mead.' And while the evidence of Old Icelandic is clearly incontrovertible,<sup>19</sup> that of Old English can hardly be regarded as "somewhat ambiguous,"<sup>20</sup> for though the digraphs *io* and *eo* were largely interchangeable, the graphs *i* and *e* were not.

The second consideration militating against the hypothesis of a PGmc. \*/i ~ e/ is that even if an *u*-umlaut of \*/e/ to \*/i/ could be assumed for North and West Germanic as a whole, it would still be impossible to postulate a complementary distribution of \*/i/ and \*/e/ in the prehistory of any of the relevant languages. For not only in North Germanic and the so-called Ingvaenic dialects, but also in

21 (1969–70), 58, 60, and *Concise Grammar*, p. 16. For a modification of Antonsen's rule to suit Gothic see M. Reis, "Phonologie des spätmittelgermanischen Vokalismus unter besonderer Berücksichtigung der Nebensilbenvokale," *Probleme der historischen Phonologie* (Wiesbaden, 1974), pp. 42–44.

15 Cf. M. S. Beeler, "Proto-Germanic [i] and [e]: One Phoneme or Two?," *Lg.*, 42 (1966), 474; Benediktsson, "Proto-Germanic Vowel System," p. 187; and Hock, "Phonemic Status," p. 328.

16 Cf. Hans Krahe and W. Meid, *Germanische Sprachwissenschaft*, 7th ed. (Berlin, 1969), II, § 62. For the peculiar OI *siau* see also Prokosch, § 99.

17 See Karl Luick, *Historische Grammatik der englischen Sprache* (Stuttgart, 1914–40), § 73, and A. Campbell, *Old English Grammar* (Oxford, 1959), § 331.3, and cf. the discussion in Benediktsson, pp. 188–90.

18 See Walther Steller, *Abriß der altfriesischen Grammatik* (Halle, 1928), § 57, and cf. OFris. *sun* 'son' beside *sunu* (–o, –a, –e). OFris. *fretho*/*frethe* (cf. OS *freðu* versus OE *fri[ð]u*–, OS *friðu*, OHG *fridu*) reflects \*/i/ > \*/e/ before \*[-a] in the genitive singular (\*/friβawez/), but OI –[f]reþr beside *friþr* (< \*/friβuz/) is best explained by lowering of \*/i/ in the second element of compound names (cf. Hock, p. 331), while OFris. *lith*, which varies with *leth*, is also inflected (like OE *liþ*) as a neuter *a*-stem (cf. Steller, §50).

19 See Beeler, pp. 473–74; Benediktsson, pp. 190–91; Antonsen, *Toward a Grammar*, p. 123; and Hock, pp. 328–29, 331. Having mentioned Beeler's objections to Marchand without comment, Panzer (*FoL*, 1 [1967], 253–54) still prefers to believe that all the Germanic languages went through a stage at which \*/i/ and \*/e/ had come to exhibit a complementary distribution.

20 Antonsen, *Toward a Grammar*, p. 123.

Old Saxon and Old High German there were positions in which the contrast  $*/i/ \neq */e/$  was preserved even after the operation of the various vocalic and consonantal influences involving the changes  $*/e/ > */i/$  and  $*/i/ > */e/$ . The relevant contexts are those in which either  $*/-e/$  or no vowel followed, since, despite persistent disagreement on the development of unaccented  $*/e/$ , the opposition  $*/i/ \neq */e/$  can be shown to have been preserved at least in originally disyllabic forms in  $*/-e/$  as well as in monosyllables.

Unaccented  $*/e/$  has been assumed to have developed according to either of two different hypotheses: (1) unaccented  $*/e/$  became  $*/i/$  in all positions after the loss of final  $*/e/$  (as well as  $*/a/$ ) and before the *i*-umlaut of  $*/e/$  to  $*/i/$ ,<sup>21</sup> (2) unaccented  $*/e/$  became  $*/i/$  only in certain positions and did not otherwise affect a preceding vowel.<sup>22</sup> Neither of these theories can be definitively proved or disproved, although the available evidence appears to favour the second rather than the first. Hock would explain OHG *ferit* 'you travel' from  $*/fariþ|ð/ < */fareþ|ð(e)/$ , its more frequent variant *faret* from a new formation *farat* (cf. *beret*, below), OS *is* 'his, its' from unstressed  $*/is/ < */es|z(a)/ < \text{PIE } */eso/$ , and its variant *es* from stressed  $*/es|z(a)/$ .<sup>23</sup> But these forms can hardly be taken to show that unaccented  $*/e/$  invariably became  $*/i/$ , since OHG *ferit* can be accounted for as a new formation after 3rd sg. *ferit* ( $*/farið(i)/ < */fareði/$ ),<sup>24</sup> OHG *faret* as a regular development of  $*/fareð(e)/$ , OS *is* as a new formation after neut. nom. *it* ( $< */it/$ ),<sup>25</sup> and OS *es* as a regular development of  $*/esa/$ . On the other hand, Antonsen would explain OHG *beret* 'you bear,' OI *falleþ* 'you fall' from  $*/bæreð(e)/$ ,  $*/falleð(e)/$ , OHG *skif* 'ship' from  $*/skipes/$  (and *skef* from  $*/skipan/$ , etc.), OS OE *wulfr*, OI *ulfr* 'wolf' from  $*/wulfes/$  (and OHG *wolf*

<sup>21</sup> See, e.g., Hock, pp. 345–46.

<sup>22</sup> See, e.g., Antonsen, *Toward a Grammar*, pp. 138–39, where unaccented  $*/e/$  is assumed to have become  $*/i/$  before final  $*/z/$  and before a high vowel (cf. Antonsen, *Concise Grammar*, pp. 15–17). For earlier suggestions see Hermann Hirt, *Handbuch des Urgermanischen* (Heidelberg, 1931), I, § 31B.1; Prokosch, § 79d.3; and Albert L. Lloyd, "Indo-European Unstressed Short *e* in Germanic," *MLN*, 76 (1961), 847–51. For a modification of Lloyd's rule, see Reis, pp. 29–32.

<sup>23</sup> See Hock, pp. 335–38.

<sup>24</sup> Cf. Wilhelm Braune and Walther Mitzka, *Althochdeutsche Grammatik*, 12th ed. (Tübingen, 1967), § 308, n. 1b: "Strittig bleibt die Erklärung der Form auf *-it*, die doch wohl analoge Neubildung ist." Also Hock, pp. 337–38, where the model "3 sg. *mahhōt* : *sagēt* : *ferit* | 2 pl. *mahhōt* : *sagēt* : *X = ferit*" is, however, rejected. But the operation of analogy would hardly be prevented by the circumstance that the verbs of the second and third weak class are "mainly derivational categories and consequently 'formes fondées'" (Hock, p. 337). Nor would the rise of a new form *ferit* be precluded by the resulting, inevitable homophony with 3rd sg. *ferit*—though such homophony was, in fact, eventually eliminated (cf. ModG. *fahrt*).

<sup>25</sup> Cf. IOHG *is* beside *es* 'of it' and see Braune-Mitzka, § 283, n. 1c.

from \*/wulfaz/, etc.).<sup>26</sup> But OHG *beret* (Alem. *berat*), OI *falleþ* cannot be accepted as conclusive evidence that unaccented \*/e/ had no effect on a preceding vowel, since they could always be accounted for as leveled from 3rd pl. *berant*, *falla* (< \*/beranð(i)/, \*/fallanð(i)/).<sup>27</sup> Nor would it be impossible to reconcile the occurrence of Goth. *-is* (which is usually derived from PGmc. \*/-esa/ < PIE \*/-eso/) with the assumption that the Old High German (and Old Saxon) genitive singular ending *-es* is an independent leveling from *des* (OS *thes*, Goth. *þis* < \*/pesa/ < \*/teso/) 'of the/that' which replaced an earlier *-as* corresponding to Ru. *-as* (OI *-s*), OE *-æs* > *-es*, OS OHG *-as* (< PGmc. \*/-asa/ < PIE \*/-oso/).<sup>28</sup>

As for the view that the only source of *a*-stem forms without *a*-umlaut is the genitive singular, it still remains to be proved definitively, since OHG *skif* and OS OE *wulf*, OI *ulfr* could always be explained as due to quasi-paradigmatic leveling—cf. OHG *skifilin* 'little ship' and OE *wylf* (< \*/wulbī/), OI *ylgr* (< \*/wulg<sup>w</sup>is/, with \*/-iz/ from the *i*-stems) 'she-wolf'—and to paradigmatic leveling from the locative singular (with \*/-e/ > \*/-ī/ > OE *-i* > *-e*) and the instrumental singular (with \*/-ō/ > OS OHG *-u*), both of which have indeed been assumed for the prehistory of North and West Germanic as a whole.<sup>29</sup> But the lack of forms with *i*-umlaut (from the locative) and *u*-umlaut (from the instrumental) in the case of *a*-stems with original \*/e/ (e.g., OS OHG *berg* 'mountain'), as well as the occurrence of derivatives with \*/-i/ (e.g., OS OHG *gibirgi* 'mountain range'), shows that leveling from both the locative (and/or instrumental) and from related forms is unlikely to be responsible for instances like OHG *skif* and OS OE *wulf*, OI *ulfr*.<sup>30</sup> Nor is it possible—irrespective of the solution adopted for final \*/e/—to explain the lack of variants in the case of *a*-stems with original \*/e/ (e.g., *\*birg* beside *berg*) by assuming that all the North and West Germanic languages originally exhibited a vocative singular form (with \*/-e/, as in Goth. *dag* versus nom.  *dags*

<sup>26</sup> See Antonsen, *Toward a Grammar*, p. 123.

<sup>27</sup> Cf. Krahe-Meid, II, § 69, and Andreas Heusler, *Altisländisches Elementarbuch*, 4th ed. (Heidelberg, 1950), § 342. The Old High German ending *-et* (with no umlaut in the stem) has also been derived from a Proto-Germanic leveled form: PIE *\*bherete/* > PGmc. *\*birið(i)/* > *\*berið/* > *\*bereð/*—see Fritz P. Knapp, "Wiederum germanisches *ē*," *PBB*, 96 (Tüb., 1974), 232.

<sup>28</sup> Cf. Hock, pp. 335–36, where the analogical explanation is accepted with regard to OHG *-es*.

<sup>29</sup> See Hock, pp. 343–44.

<sup>30</sup> The case of an *r*-stem with original \*/e/ like OI *syster* 'sister' is of course different, since \*/i/ was here leveled from the nominative plural: *\*swesterez/* > *\*swestiriz/* > *\*swist(i)riz/* > *systr* versus OE *sweostor*, OFris. *swester/sister*, OS *swestar*, OHG *swester/swister* (Goth. *swistar* < *\*swestēr/*).

'day'), and that "the subsequent syncretism of nominative singulars like *\*spek(R)*, *\*wolf(R)*, *\*berg(R)* with vocative singulars like *\*spik*, *\*wulf*, *\*berg*" produced "the doublets *\*spek(R)/spik(R)* and *\*wolf(R)/wulf(R)* [as opposed to] *\*berg(R)*."<sup>31</sup> For on the assumption of a distinct vocative form for North and West Germanic, the available evidence appears to justify only the postulation of a subsequent functional replacement of the vocative by the nominative, and not of a formal syncretism resulting in "the new category nominative/vocative."<sup>32</sup>

The indirect evidence of the *a*-stems with original *\*e/* would therefore seem to show that instances like OHG *skif* 'ship,' OE *spic* (beside *spec* < *\*spikan/*) 'bacon,' and OS OE *wulf*, OI *ulfr* 'wolf' can be explained analogically only if it is assumed that they were leveled from the genitive singular<sup>33</sup> and, consequently, that unaccented *\*e/* had no effect on the preceding vowel. But even this conclusion must remain a tentative one, since analogical replacement alone cannot have been responsible either for the extreme scarcity of forms with */e/* by *a*-umlaut of *\*i/* or for the abundance of forms with */u/* instead of, or beside, expected */o/*, especially in certain dialects. In the former case, it would appear that the assimilatory trend to lower *\*i/* to identity with *\*e/*—being less powerful than the tendency *\*e/* > *\*i/* before *\*[i]*—was resisted, with varying results, in all the North and West Germanic languages:<sup>34</sup> OHG *skef/skif*, OS OFris. *skip*, OE *sċip*, OI (Goth.) *skip* 'ship,' OHG OFris. *spek*, OE *spec/spic*, OI *spik* 'bacon,' etc. In the latter case, the change *\*u/* > *\*[o]* (> *\*o/*) before *\*[a]* appears to have been prevented, at least within certain dialects, by a neighbouring labial—cf. OE *full*, OFris. OS *ful/fol*, OHG *fol*, OI *fullr* (Goth. *fulls* < *\*fullaz/*) 'full,' OE *fugol*, OFris. *fugel*, OS OHG *fugal/fogal*, OI *fugl/fogl* (Goth. *fugls* < *\*fuglaz/*) 'fowl, bird'—so that there is always the possibility of an occasional extension of the idiolectal variation between *\*u/* and *\*o/* in labial contexts to other environments, e.g., in OI *gull/goll* beside OE OFris. OS OHG *gold* (Goth. *gulþ* < *\*gulþan/*) 'gold.'

In the absence of evidence to the contrary, it would nevertheless seem reasonable to conclude that the change of unaccented *\*e/* to *\*i/* should be assumed to have occurred in all positions in pre-Gothic, and only in favourable contexts in the prehistory of the individual North and West Germanic languages. Here, the tendency of unac-

<sup>31</sup> Hock, p. 345.

<sup>32</sup> Hock, p. 345.

<sup>33</sup> Cf. OFris. *fretho*, OS *freðu* among the *u*-stems (n. 18, above).

<sup>34</sup> Cf. Cercignani, *Lg.*, 56 (1980), § 2.

cented \*/e/ to become \*/i/ should be postulated before high vowels, as in OHG *henin* (< \*/hanin(i)/ < \*/haneni/)<sup>35</sup> 'cock (dat.)' beside leveled *hanin*<sup>36</sup> and OHG *lembir* 'lambs' (with \*[-u] representing PGmc. \*/-ō/, \*/lambezō/),<sup>37</sup> before nasals (at least when followed by another consonant), as in OS *penning*, OHG *pfenning* (< \*/pan-ðengaz/) 'penny,' and before final \*/z/,<sup>38</sup> as in OE OS *men* (< \*/mannez/) beside OS OHG *man* (< /mannaz/) 'men.'<sup>39</sup> Forms like OHG *elina* 'ell' and *mihhil* (OS *mikil*) 'great' may well have PGmc. \*/-inō/ < PIE \*/-inā/ (cf. Goth. *aleina*, with \*/-inō/ < \*/-inā/) and PGmc. \*/-ila-/ < PIE \*/-ilo-/ rather than PIE \*/-enā/ and \*/-elo-/,<sup>40</sup> and cannot therefore be adduced to show that \*/e/ in unaccented syllables became \*/i/ "in viel größerem Umfang als in den Ton-silben."<sup>41</sup> Nor should it be forgotten that participial formations in \*/-ina-/ (e.g., OE *cymen* 'come') can be derived from PIE \*/-ino-/ rather than from \*/-eno-/,<sup>42</sup> and that \*/i/ from \*/e/ before a single nasal would, at any rate, be quite likely in the latter.<sup>43</sup>

If the change of unaccented \*/e/ to \*/i/ in North and West Germanic occurred only in favourable contexts, then the contrast \*/i/ ≠

<sup>35</sup> For the use of [h] to designate a voiceless velar spirant on the model of the symbol [g] denoting its voiced counterpart see Fausto Cercignani, *The Consonants of German: Synchrony and Diachrony* (Milan, 1979), § 1.

<sup>36</sup> Cf. Braune-Mitzka, § 221, n. 2. Lloyd (*MLN*, 76 [1961], 849–50) would explain the dative variant *hanen* as having unaccented *e* from the genitive form *hanen* (< \*/hanenaz/), which occurs beside *hanin*.

<sup>37</sup> Knapp (*PBB*, 96 [Tüb., 1974], 222) objects that an *u*-umlaut of \*/e/ to \*/i/ cannot be ascribed to Proto-Germanic even in the case of accented syllables, and that in Old Saxon and Old High German \*/e/ > \*/i/ occurs only before retained \*[-u], e.g., in *giðu* and *gibu* 'I give.' But while the change in question should hardly be assumed for Proto-Germanic, unaccented vowels cannot be expected to behave (as Knapp himself notes) exactly like accented ones, and \*[-u] would of course be lost earlier in third than in second syllables.

<sup>38</sup> Cf. the Old Icelandic 'R-umlaut,' as in *ker* (Goth. *kas* < \*/kazan/) 'vessel,' and see further Adolf Noreen, *Altisländische und altnorwegische Grammatik*, 4th ed. (Halle, 1923), § 71.

<sup>39</sup> Cf. also OI *fōtr*, OE *fētr*/fēt, OFris. *fēt* (< \*/fōtez/).

<sup>40</sup> Cf. H. Krahe and W. Meid, *Germanische Sprachwissenschaft* (Berlin, 1967), III, §§ 87a.2, 94.1b, and Lloyd, pp. 848–49.

<sup>41</sup> H. Krahe and W. Meid, *Germanische Sprachwissenschaft*, 7th ed. (Berlin, 1969), I, § 45.

<sup>42</sup> For the latter see, e.g., Campbell, § 331.2, n. 1. Antonsen (*Concise Grammar*, p. 22) assumes PIE \*/-yno-/ , despite his recognition that the dative singular of the Germanic *i*-stems (PIE \*/-eji/ and \*/-oji/) indicates that \*/[i] had already acquired phonemic status in (late) Proto-Indo-European. Be that as it may, the phonemization of PIE \*/[i] and \*/[u] can hardly have occurred, as will be argued in a separate article, "through 'system balance' upon the rise of \*/[i] and \*/[ū]" (Antonsen, *Toward a Grammar*, p. 138).

<sup>43</sup> Cf. the tendency \*/e/ > \*/i/ in accented syllables, as in OE *niman*, OFris. *nima/nema*, OS *niman/neman* (OHG *neman*, OI *nema* and Goth. *niman* < \*/nemanan/) 'take.'

*\*/e/* was obviously preserved also in unaccented syllables throughout the prehistory of all the relevant languages, as can be seen from original instances like *\*/mari/* 'sea' — *\*/fare/* 'travel thou' and *\*/fareði/* 'he travels' — */fareðe/* 'you travel,' as well as from later forms like *\*/mari/* 'sea' — *\*/fare/* (< *\*/farē/* < *\*/fara¹/*) 'journey (dat. sg.)' and *\*/farið/* 'he travels' — *\*/fareð/* 'you travel.'<sup>44</sup> But regardless of the solution adopted for the development of unaccented *\*/e/*, the fact remains that in North and West Germanic the contrast *\*/i/* ≠ *\*/e/* was preserved at least in originally disyllabic forms in *\*/-e/* as well as in monosyllables. Forms like OE *mere*, OS OHG *meri* (< *\*/mari/*) 'sea' and OE OHG *far*<sup>45</sup> (< *\*/fare/*) 'travel thou' cannot of course be taken to show "the loss of final *-e* before weakly stressed *-e* became *-i*."<sup>46</sup> But their evidence is of paramount importance, since what they do show is that final *\*/e/* preserved its separate identity, and that no *i*-umlaut occurred in original disyllables in *\*/-e/-cf.* OE *fær[e]st*,<sup>47</sup> OHG *feris[t]* (< *\*/faris(i)/* < *\*/faresi/*) 'thou travelst.' This is confirmed by cases like OE (Nhb) *gē-wit*<sup>48</sup> (< *\*/wite/*) 'know thou' and OE *ber* (< *\*/ðere/*) 'bear thou'—cf. OE *bir[e]st* (< *\*/beris(i)* < *\*/beresi/*) 'thou bearest'—which also prove beyond reasonable doubt that in originally disyllabic forms in *\*/-e/* accented *\*/e/* was preserved<sup>49</sup> and, consequently, that the opposition *\*/i/* ≠ *\*/e/* was retained at least here. Comparable forms are, therefore, OI *vit* and *ber* (despite 2nd sg. ind. *berr*), as well as OS *ber* beside analogical *bir*,<sup>50</sup> while a reconstructed minimal pair would be *\*/wige/* 'kill thou' ≠ *\*/wege/* 'carry thou' or, according to the first hypothesis discussed above, *\*/wig/* ≠ *\*/weg/*.<sup>51</sup>

As for the suggestion that "the opposition *i : e* may have maintained its distinctive function in Proto-Germanic before *ē* and, at any rate,

<sup>44</sup> The claim that the developments in unaccented syllables establish a contrast *\*/i/* ≠ *\*/e/* for late Proto-Germanic (see Reis, p. 24) implies the dubious postulation of exactly the same changes not only for North and West Germanic as a whole, but also for pre-Gothic.

<sup>45</sup> OE *far* for *\*/fær*, after *faran*—cf. OE *fære* < *\*/fare/* 'journey (dat. sg.)' above.

<sup>46</sup> W. P. Lehmann, "A Definition of Proto-Germanic," *Lg.*, 37 (1961), 71, and cf. Benediktsson, p. 193. OE *mec* (ONorw. *mek*) 'me (acc.)' which has been adduced on both sides as representing PGmc. *\*/meke/* (cf. Lloyd, p. 851), goes back to PGmc. *\*/mek/* (see below) and is therefore irrelevant.

<sup>47</sup> With *i*-umlaut of analogical *\*/a/* (as in *faran*) instead of *\*/æ/-cf.* Campbell, § 193c, and Karl Brunner, *Altenglische Grammatik*, 3rd ed. (Tübingen, 1965), § 96.3.

<sup>48</sup> Beside OE *wite*, a subjunctive form (cf. OHG *wizzis[t]*).

<sup>49</sup> Cf. Benediktsson, p. 193: "If the pre-Old English form had been *\*bir*, we should not, in view of 2nd sg. ind. *bires(t)*, have expected analogical replacement of *i* by *e*."

<sup>50</sup> Like OHG *bir*, the Old Saxon variant should of course be explained after *biris[t]* (cf. Krahe-Meid, II, § 79).

<sup>51</sup> Cf. Benediktsson, p. 193, and Hock, p. 331.

before *ai*,<sup>52</sup> it should be observed that OI *stige* (beside *stege*) ‘ladder’—as well as OSw. *bughi* (beside *boghi*) ‘bow’—can be explained as having retained the original vowel before an intervening \*/j/ (\*/\*stijjōn/ > \*/stijja/ > *stigi* > -e)<sup>53</sup> rather than before \*/-ēn/ and, further, that Ru. *wurte* (OI *orte* < \*/wurhtēð/) ‘wrought’ may have graphic *u* for *o*, and so may Ru. *-kurne* (OI *korne* < \*/kurna<sup>i</sup>/) ‘corn (dat.)’ in the same inscription.<sup>54</sup> Again, instances like OHG *bibēn* (but eModG *beben*) ‘tremble’ can be explained as having \*/i/ from original forms with \*/-j-/ (cf. OE *bifan*), while instances like OHG *lebēn* (OE *libban*, *lifgan*, OS *libbian*) ‘live’—as well as OHG *wonēn* (OE *wunian*) ‘dwell’—need not reflect the vowel of older syncopated preterites, e.g., \*/liððōn/ > OHG *\*lepta* ‘lived.’<sup>55</sup> Nor is it necessary, in the case of OI *dōtter*, OE *dohtor*, OFris. *dochter*, OS *dohtar*, OHG *tohter* (Goth. *daūhtar* < \*/ðuhtēr/) ‘daughter’ either to have recourse to leveling from a form like gen. sg. \*/ðuhtraz/<sup>56</sup> or to devise “a Proto-Germanic change of \*/e/ to *a* [before] *r*.”<sup>57</sup>

With regard to monosyllables the evidence is obviously rather scanty, but \*/e/ was clearly preserved in, e.g., \*/sehs/ ‘six’—cf. OI *sex*, OE *se[o]x* > *si[e]x*, OFris. *sex*, OS OHG *sehs* (Goth. *saihs*)—which would contrast not only with forms containing original \*/i/, such as \*/it/ ‘it’—cf. Ru. *[i]t* (OI *pat*), OE *hit*, OFris. *hit* > *het*, OS *it*, OHG *iz* > *ez* (Goth. *ita*)—but also with those in which \*/i/ represented \*/e/, such as \*/ik/ ‘I’ > Ru. *ik* (beside Ru. OI *ek*), OE *iċ*, OFris. OS *ik*, OHG *ih* (Goth. *ik* < \*/ek/). The vowel of \*/ik/ can of course be explained as leveled from the originally unstressed dat. \*/miz/ (OFris. *mĭ*, OS *mĭ*, OHG *mir*) < stressed \*/mez/ (OI *mēr*, OE *mē* and Goth. *mis*)<sup>58</sup> or, according to the first hypothesis discussed above, as developed from \*/e/ in unstressed \*/ek/.<sup>59</sup> The same applies to the ac-

<sup>52</sup> Benediktsson, p. 192 and pp. 191–92, followed by Reis, pp. 24–25. The \*/i/ in the past participle of the first strong class (OHG *ritan* ‘ride’—*reit*, *ritum*—*giritan*) should be explained as reflecting the combined influence of the preterite plural and of the *i*-timbre of the present—cf. Prokosch, § 41a, and Hock, pp. 340–41. In view of the evidence of Gothic, the resulting contrast \*/i/ ≠ \*/e/ before \*/-a/ cannot of course be regarded as late Proto-Germanic (see Reis, p. 25) but, at most, as ‘Northwest Germanic.’

<sup>53</sup> See Nils Lid, “Den nordiske nominativ singularis av maskuline *an*-stammer,” *NTS*, 16 (1952), 237–40, and cf. Antonsen, *Concise Grammar*, p. 21.

<sup>54</sup> Cf. Antonsen, *Concise Grammar*, pp. 5, 13.

<sup>55</sup> Cf. Marchand, *Lg.*, 33 (1957), 348, and Hock, pp. 332–33.

<sup>56</sup> See Benediktsson, pp. 191–92.

<sup>57</sup> Hock, pp. 333–34. On Ru. *swestar* reformed according to acc. \*/swestarun/ see Antonsen, *Concise Grammar*, p. 40, against Krause, § 28.6.

<sup>58</sup> Cf. Antonsen, *Concise Grammar*, pp. 22, 40, where Ru. *mez* (or, traditionally, *meR*), is interpreted as *mēz*, and OE *mē* derived from \*/mēz/ rather than from *me*—but see Krause, § 24, and Krahe-Meid, II, § 32.

<sup>59</sup> See, e.g., Benediktsson, p. 194, and Hock, p. 345.

cusative form \*/mik/ (OI OS *mik*, OHG *mih*) beside \*/mek/ (ONorw. *mek*, OE *meċ* and Goth. *mik*), which is from PIE \*/me/, with \*/k/ after \*/ek/,<sup>60</sup> and not, as is often assumed, from a PGmc. \*/meke/ representing a PIE \*/mege/.<sup>61</sup>

Thus, even if the early changes \*/e/ > \*/i/ (including \*/e/ > \*/i/ before \*/[–u]) and \*/i/ > \*/e/ could be assumed for all the Germanic languages, it would still be impossible to ascribe \*/i ~ e/ to any stage of Proto-Germanic, since the opposition \*/i/ ≠ \*/e/ can be shown to have been preserved throughout the prehistory of all the North and West Germanic dialects at least in originally disyllabic forms in \*/–e/ as well as in monosyllables. In pre-Saxon and pre-High German, where \*/e/ was actually identified with \*/i/ also before \*/[–u], the functional load of \*/i/ ≠ \*/e/ obviously came to be lower than in the rest of North and West Germanic, irrespective of whether or not the raising of \*/e/ to \*/i/ in the various positions occurred at different times. But the contrast survived even in these dialects, and its functional load was subsequently increased, as in the rest of North and West Germanic, through paradigmatic levelings and other analogical processes, as well as through the gradual loss of vocalic determinants.

<sup>60</sup> Cf. Krahe-Meid, II, § 32.

<sup>61</sup> See, e.g., *Oxford Dictionary of English Etymology*, ed. C. T. Onions et al. (Oxford, 1966), s.v. *me*.